Digital Verification diploma

Project 1

FIFO

Name: Taqi alrahman Ahmed Mohamed

Mail: taqi26ahmed@gmail.com

Contents

No table of contents entries found.

Verification Plan

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
FIFO_1	When the reset is asserted, the fifo should be cleared and internal pointers should be reset, counter should be reset,dataout should be 0 and flags should be reset to their default values	Directed at the start of the simulation	-	assertions and golden model at scoreboard
FIFO_2	when reset is disabled and wr_en = 1, r then data_in should be inserted to the fifo If fifo isnt full	Directed loop with randomized data in with reset = 0, wr_en = 1,rd_en=0,		golden model at scoreboard
FIFO_3	when reset is disabled and wr_en = 1, ; wr_ack should be high,	Directed loop with randomized data in with reset = 0, wr_en = 1,rd_en=0,	cover cross coverage of wr_en,rd_en, wr_ack bins {0,X,1} should be illegal	concurrent ssertion for wr_ack
FIFO_4	when reset is disabled full flag should be high if full,	Directed loop with randomized data in with reset = 0, wr_en = 1,rd_en=0,	cover cross coverage of wr_en,rd_en,full bins {X,1,1} should be illegal	concurrent assertion for full flag
FIFO_5	when reset is disabled almostfull flag should be high if full after one write,	Directed loop with randomized data in with reset = 0, wr_en = 1,rd_en=0,	cover cross coverage of wr_en,rd_en,almostfull	concurrent assertion for almost full
FIFO_6	when reset is disabled wr_en = 1 and is full Overflow should be high	Directed loop with randomized data in with reset = 0, wr_en = 1,rd_en=0,	cover cross coverage of wr_en,rd_en,overflow bins {0,X,1} should be illegal	concurrent assertion for overflow
FIFO_7	when reset is disabled and rd_en = 1, then data_out should be the first value in the front of the fifo if not empty	Directed loop with randomized data in with reset = 0, wr_en = 0,rd_en=1,		golden model at scoreboard

FIFO_8	when reset is disabled empty flag should be high if empty,	Directed loop with randomized data in with reset = 0, wr_en = 0,rd_en=1,	cover cross coverage of wr_en,rd_en,empty	concurrent assertion for empty flag
FIFO_9	when reset is disabled almostempty flag should be high if empty after one read,	Directed loop with randomized data in with reset = 0, wr_en = 0,rd_en=1,	cover cross coverage of wr_en,rd_en,almostempty	concurrent assertion for almost empty
FIFO_10	when reset is disabled rd_en = 1 and is empty Underflow should be high	Directed loop with randomized data in with reset = 0, wr_en = 0,rd_en=1,	cover cross coverage of wr_en,rd_en,underflow bins {X,0,1} should be illegal	concurrent assertion for undeflow
FIFO_11	write and read pointers should stay in valid range and wraparound	loop with randomized input under constraints	cover cross coverage of wr_en,rd_en, wr_ack	golden model at scoreboard
FIFO_12	when reset is disabled and rd_en = 1, wr_en = 1 and full should allow writing	loop with randomized input under constraints	-	golden model at scoreboard
FIFO_13	when reset is disabled and rd_en = 1, wr_en = 1 and empty should allow writing	loop with randomized input under constraints	-	golden model at scoreboard
FIFO_14	should allow different orders of reading and writing	loop with randomized input under constraints	-	golden model at scoreboard

Design Bugs:

1- Missing resets for some output signals

```
always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
    if (!fifoif.rst_n) begin
        wr_ptr <= 0;
        fifoif.wr_ack <= 0;      //BUG: reset wr_ack was missing
        fifoif.overflow <= 0;    //BUG: reset for overflow was missing
```

```
always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
    if (!fifoif.rst_n) begin
        rd_ptr <= 0;
        fifoif.data_out <= 0; //BUG: reset for dataout was missing
        fifoif.underflow <= 0; //BUG: reset for underflow was missing
    end
```

2- Overflow wasn't pulled down to 0 in successful writes

```
    else if (fifoif.wr_en && count < fifoif.FIFO_DEPTH) begin
        mem[wr_ptr] <= fifoif.data_in;
        fifoif.wr_ack <= 1;
        wr_ptr <= wr_ptr + 1;
        fifoif.overflow <= 0;  //BUG: should pull overflow back to 0 if we wrote normally
    end
    else begin
```

3- Underflow was driven as a combinational output

```
//assign fifoif.underflow = (fifoif.empty && fifoif.rd_en)? 1 : 0;
//BUG: underflow should be driven as a sequential signal
```

Fix:

```
    else if (fifoif.rd_en && count != 0) begin
        fifoif.data_out <= mem[rd_ptr];
        rd_ptr <= rd_ptr + 1;
        fifoif.underflow<= 0; //BUG: should pull underflow back to 0 if we read normally
    end
    else begin  //BUG: handling of underflow was missing
        if (fifoif.empty & fifoif.rd_en)
            fifoif.underflow <= 1;
        else
            fifoif.underflow <= 0;
    end
```

4- Didn't handle count when both read and write were high but the fifo was empty or full

```
        end
        else begin
            if  ( ({fifoif.wr_en, fifoif.rd_en} == 2'b10) && !fifoif.full)
                count <= count + 1;
            else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b01) && !fifoif.empty)
                count <= count - 1;
            //BUG: should check when both are high and increment counter if empty
            else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b11) && fifoif.empty)
                count <= count + 1;
            //BUG: should check when both are high and decrement counter based if full
            else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b11) && fifoif.full)
                count <= count - 1;
        end
    end
```

5- Almost full flag condtion was wrong

Fix:

```
assign fifoif.almostfull = (count == fifoif.FIFO_DEPTH-1)? 1 : 0; //BUG: was
written as fifodepth -2 instead of 1
```

Codes

Fixed Design with assertions

```
////////////////////////////////////////////////////////////////////////////////
// Author: Kareem Waseem
// Course: Digital Verification using SV & UVM
//
// Description: FIFO Design
//
////////////////////////////////////////////////////////////////////////////////
module FIFO(FIFO_if.DUT fifoif);


localparam max_fifo_addr = $clog2(fifoif.FIFO_DEPTH);

reg [fifoif.FIFO_WIDTH-1:0] mem [fifoif.FIFO_DEPTH-1:0];

reg [max_fifo_addr-1:0] wr_ptr, rd_ptr;
reg [max_fifo_addr:0] count;

always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
    if (!fifoif.rst_n) begin
        wr_ptr <= 0;
        fifoif.wr_ack <= 0;     //BUG: reset wr_ack was missing
        fifoif.overflow <= 0;   //BUG: reset for overflow was missing
    end
    else if (fifoif.wr_en && count < fifoif.FIFO_DEPTH) begin
        mem[wr_ptr] <= fifoif.data_in;
        fifoif.wr_ack <= 1;
        wr_ptr <= wr_ptr + 1;
        fifoif.overflow <= 0;  //BUG: should pull overflow back to 0 if we wrote normally
    end
    else begin
        fifoif.wr_ack <= 0;
        if (fifoif.full & fifoif.wr_en)
            fifoif.overflow <= 1;
        else
            fifoif.overflow <= 0;
    end
end

always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
    if (!fifoif.rst_n) begin
        rd_ptr <= 0;
        fifoif.data_out <= 0; //BUG: reset for dataout was missing
        fifoif.underflow <= 0; //BUG: reset for underflow was missing
    end
    else if (fifoif.rd_en && count != 0) begin
        fifoif.data_out <= mem[rd_ptr];
        rd_ptr <= rd_ptr + 1;
        fifoif.underflow<= 0; //BUG: should pull underflow back to 0 if we read normally
```

```
    end
    else begin  //BUG: handling of underflow was missing
        if (fifoif.empty & fifoif.rd_en)
            fifoif.underflow <= 1;
        else
            fifoif.underflow <= 0;
    end
end

always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
    if (!fifoif.rst_n) begin
        count <= 0;

    end
    else begin
        if  ( ({fifoif.wr_en, fifoif.rd_en} == 2'b10) && !fifoif.full)
            count <= count + 1;
        else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b01) && !fifoif.empty)
            count <= count - 1;
        //BUG: should check when both are high and increment counter if empty
        else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b11) && fifoif.empty)
            count <= count + 1;
        //BUG: should check when both are high and decrement counter based if full
        else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b11) && fifoif.full)
            count <= count - 1;
    end
end

assign fifoif.full = (count == fifoif.FIFO_DEPTH)? 1 : 0;
assign fifoif.empty = (count == 0)? 1 : 0;
//assign fifoif.underflow = (fifoif.empty && fifoif.rd_en)? 1 : 0; //BUG: underflow
should be driven as a sequential signal
assign fifoif.almostfull = (count == fifoif.FIFO_DEPTH-1)? 1 : 0; //BUG: was written as
fifodepth -2 instead of 1
assign fifoif.almostempty = (count == 1)? 1 : 0;

//=================== ASSERTIONS ==========================
    always_comb begin
        if(!fifoif.rst_n) begin
            assert final( (wr_ptr == 'b0) && (rd_ptr == 'b0)
                            && (count == 'b0));
        end
    end

    property wr_ack_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (fifoif.wr_en && !fifoif.full)
|=> fifoif.wr_ack ;
    endproperty
    property overflow_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (fifoif.wr_en && fifoif.full)
|=> fifoif.overflow ;
    endproperty
    property underflow_pr;
```

```
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (fifoif.rd_en && fifoif.empty)
|=> fifoif.underflow ;
    endproperty
    property empty_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (count == 'b0) |-> fifoif.empty;
    endproperty
    property full_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (count == fifoif.FIFO_DEPTH) |->
fifoif.full;
    endproperty
    property almost_full_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (count == fifoif.FIFO_DEPTH -1)
|-> fifoif.almostfull;
    endproperty
    property almost_empty_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) (count == 1) |->
fifoif.almostempty;
    endproperty
    property wr_ptr_wraparound_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) ((wr_ptr == fifoif.FIFO_DEPTH-1)
&& fifoif.wr_en && !fifoif.full) |=> wr_ptr ==0 ;
    endproperty
    property ptr_threshold_pr;
        @(posedge fifoif.clk) disable iff(!fifoif.rst_n) ((rd_ptr < fifoif.FIFO_DEPTH) &&
(wr_ptr<fifoif.FIFO_DEPTH) && (count<=fifoif.FIFO_DEPTH));
    endproperty

    wr_ack_asrt: assert property (wr_ack_pr)
        else $error("Assertion wr_ack_asrt failed!");
    overflow_asrt: assert property (overflow_pr)
        else $error("Assertion overflow_asrt failed!");
    underflow_asrt: assert property (underflow_pr)
        else $error("Assertion underflow_asrt failed!");
    empty_asrt: assert property (empty_pr)
        else $error("Assertion empty_asrt failed!");
    full_asrt: assert property (full_pr)
        else $error("Assertion full_asrt failed!");
    almost_full_asrt: assert property (almost_full_pr)
        else $error("Assertion almost_full_asrt failed!");
    almost_empty_asrt: assert property (almost_empty_pr)
        else $error("Assertion almost_empty_asrt failed!");
    wr_ptr_wraparound_asrt: assert property (wr_ptr_wraparound_pr)
        else $error("Assertion wr_ptr_wraparound_asrt failed!");
    ptr_threshold_asrt: assert property (ptr_threshold_pr)
        else $error("Assertion ptr_threshold_asrt failed!");

    wr_ack_cov: cover property (wr_ack_pr);
    overflow_cov: cover property (overflow_pr);
    underflow_cov: cover property (underflow_pr);
    empty_cov: cover property (empty_pr);
    full_cov: cover property (full_pr);
    almost_full_cov: cover property (almost_full_pr);
    almost_empty_cov: cover property (almost_empty_pr);
    wr_ptr_wraparound_cov: cover property (wr_ptr_wraparound_pr);
```

```
    ptr_threshold_cov: cover property (ptr_threshold_pr);

endmodule
```

Interface

```
interface FIFO_if(clk);
    parameter FIFO_WIDTH = 16;
    parameter FIFO_DEPTH = 8;

    input clk;

    logic [FIFO_WIDTH-1:0] data_in;
    logic rst_n, wr_en, rd_en;

    logic [FIFO_WIDTH-1:0] data_out;
    logic wr_ack,full, empty, almostfull, almostempty, underflow, overflow;

    event ev_trig;

    modport DUT (
    input data_in,clk,rst_n, wr_en, rd_en,
    output data_out,wr_ack, overflow, full, empty, almostfull, almostempty, underflow
    );

    modport MONITOR (
    input ev_trig,data_in,clk,rst_n, wr_en, rd_en,data_out,wr_ack, overflow, full, empty,
almostfull, almostempty, underflow
    );

    modport TEST (
    input clk,data_out,wr_ack, overflow, full, empty, almostfull, almostempty, underflow,
    output ev_trig,data_in,rst_n, wr_en,rd_en
    );

endinterface
```

Transaction package

```
package FIFO_transaction_pkg;
    parameter FIFO_WIDTH = 16;
    parameter FIFO_DEPTH = 8;

    class FIFO_transaction;

        rand logic [FIFO_WIDTH-1:0] data_in;
        rand logic rst_n, wr_en, rd_en;
        logic [FIFO_WIDTH-1:0] data_out;
        logic wr_ack, overflow;
        logic full, empty, almostfull, almostempty, underflow;

        integer RD_EN_ON_DIST;
        integer WR_EN_ON_DIST;

        function  new(integer RD_EN_ON_DIST = 30, integer WR_EN_ON_DIST = 70);
            this.RD_EN_ON_DIST = RD_EN_ON_DIST;
            this.WR_EN_ON_DIST = WR_EN_ON_DIST;
        endfunction

        constraint reset_c{
            rst_n dist {1:= 98, 0:=2};
        }

        constraint wr_en_c{
            wr_en dist { 1:= WR_EN_ON_DIST , 0:= 100-WR_EN_ON_DIST};
        }

        constraint rd_en_c{
            rd_en dist { 1:= RD_EN_ON_DIST , 0:= 100-RD_EN_ON_DIST};
        }

    endclass
endpackage
```

Shared package

```
package shared_pkg;
    bit test_finished;
    integer error_count = 0;
    integer correct_count = 0;
endpackage
```

Scoreboard package

```
package FIFO_scoreboard_pkg;
    import FIFO_transaction_pkg::*;
    import shared_pkg::*;
    class FIFO_scoreboard ;

        logic [FIFO_WIDTH-1:0] data_out_ref;
        logic wr_ack_ref, overflow_ref, full_ref, empty_ref, almostfull_ref,
almostempty_ref, underflow_ref;

        logic [FIFO_WIDTH-1:0] myQueue[$];

        task check_data(FIFO_transaction f_txn);
            ref_model(f_txn);
            if(data_out_ref !== f_txn.data_out)begin
                $display("ERROR: data_out incorrect, data_in=%0h, rd_en=%0h, wr_en=%0h,
dut_data_out=%0h, ref_data_out = %0h",
                            f_txn.data_in, f_txn.rd_en,
f_txn.wr_en,f_txn.data_out,data_out_ref);
                error_count++;
            end
            else begin
                correct_count++;
            end
        endtask

        task ref_model(FIFO_transaction f_txn);
            if(!f_txn.rst_n) begin
                empty_ref = 1;
                full_ref =0;
                almostempty_ref = 0;
                almostfull_ref = 0;
                data_out_ref = 0;
                wr_ack_ref = 0;
                overflow_ref = 0;
                underflow_ref = 0;

                myQueue.delete();
            end

            else begin

                if(f_txn.rd_en && f_txn.wr_en) begin
                    if(myQueue.size() == 0) begin //allow write only
                        myQueue.push_back(f_txn.data_in);
                        wr_ack_ref = 1;

                    end
                    else if(myQueue.size() == FIFO_DEPTH) begin //allow read only
                        data_out_ref = myQueue.pop_front();
                        wr_ack_ref = 0;
                    end
```

```
                    else begin //one read and one write
                        myQueue.push_back(f_txn.data_in);
                        wr_ack_ref = 1;
                        data_out_ref = myQueue.pop_front();

                    end
                end
                else if(f_txn.rd_en)begin //read
                    if(myQueue.size() != 0)begin
                        data_out_ref = myQueue.pop_front();
                        wr_ack_ref = 0;
                    end
                end
                else if(f_txn.wr_en)begin //write
                    if(myQueue.size() != FIFO_DEPTH)begin
                        myQueue.push_back(f_txn.data_in);
                        wr_ack_ref = 1;
                    end
                end

                empty_ref = (myQueue.size() == 0)? 1:0;
                almostempty_ref = (myQueue.size() == 1)? 1:0;
                full_ref = (myQueue.size() == FIFO_DEPTH)? 1:0;
                almostfull_ref = (myQueue.size() == FIFO_DEPTH -1)? 1:0;
            end
        endtask

    endclass
endpackage
```

Coverage package

```
package FIFO_coverage_pkg;
    import FIFO_transaction_pkg::*;

    class FIFO_coverage;
        FIFO_transaction F_cvg_txn;

        covergroup cov_group;
            // option.per_instance = 1;
            wr_en_cp: coverpoint F_cvg_txn.wr_en;
            rd_en_cp: coverpoint F_cvg_txn.rd_en;
            wr_ack_cp: coverpoint F_cvg_txn.wr_ack;
            fullcp: coverpoint F_cvg_txn.full;
            almostfull_cp: coverpoint F_cvg_txn.almostfull;
            empty_cp: coverpoint F_cvg_txn.empty;
            almostempty_cp: coverpoint F_cvg_txn.almostempty;
            overflowcp: coverpoint F_cvg_txn.overflow;
            underflow_cp: coverpoint F_cvg_txn.underflow;

            wr_rd_en_wr_ack_cp :cross wr_en_cp,rd_en_cp,wr_ack_cp{
                illegal_bins wr_en0_rd_en0_wr_ack1 = binsof(wr_en_cp) intersect {0} &&
                                                     binsof(rd_en_cp) intersect {0} &&
                                                     binsof(wr_ack_cp) intersect {1} ;
                illegal_bins wr_en0_rd_en1_wr_ack1 = binsof(wr_en_cp) intersect {0} &&
                                                     binsof(rd_en_cp) intersect {1} &&
                                                     binsof(wr_ack_cp) intersect {1} ;
            }
            wr_rd_en_full_cp :cross wr_en_cp,rd_en_cp,fullcp{
                illegal_bins wr_en0_rd_en1_full1 = binsof(wr_en_cp) intersect {1} &&
                                                     binsof(rd_en_cp) intersect {1} &&
                                                     binsof(fullcp) intersect {1} ;
                illegal_bins wr_en1_rd_en1_full1 = binsof(wr_en_cp) intersect {0} &&
                                                     binsof(rd_en_cp) intersect {1} &&
                                                     binsof(fullcp) intersect {1} ;
            }
            wr_rd_en_empty_cp :cross wr_en_cp,rd_en_cp,empty_cp;
            wr_rd_en_almostfull_cp :cross wr_en_cp,rd_en_cp,almostfull_cp;
            wr_rd_en_almostempty_cp :cross wr_en_cp,rd_en_cp,almostempty_cp;
            wr_rd_en_overflow_cp :cross wr_en_cp,rd_en_cp,overflowcp{
                illegal_bins wr_en0_rd_en0_ov1 = binsof(wr_en_cp) intersect {0} &&
                                                     binsof(rd_en_cp) intersect {0} &&
                                                     binsof(overflowcp) intersect {1} ;
                illegal_bins wr_en0_rd_en1_ov1 = binsof(wr_en_cp) intersect {0} &&
                                                     binsof(rd_en_cp) intersect {1} &&
                                                     binsof(overflowcp) intersect {1} ;
            }
            wr_rd_en_underflow_cp :cross wr_en_cp,rd_en_cp,underflow_cp{
                illegal_bins wr_en0_rd_en0_uf1 = binsof(wr_en_cp) intersect {0} &&
                                                     binsof(rd_en_cp) intersect {0} &&
                                                     binsof(underflow_cp) intersect {1} ;
                illegal_bins wr_en1_rd_en0_uf1 = binsof(wr_en_cp) intersect {1} &&
                                                     binsof(rd_en_cp) intersect {0} &&
```

```
                                                    binsof(underflow_cp) intersect {1} ;
            }

        endgroup

        function new();
            cov_group = new();
        endfunction //new()

        function void sample_data(FIFO_transaction F_txn);
            F_cvg_txn = F_txn;
            cov_group.sample();
        endfunction

    endclass //FIFO_coverage

endpackage
```

NOTE:

I made some illegal bins for:

1- **wr_en_cp,rd_en_cp,overflow_cp**
   (it is impossible for overflow to be high when there is no write operation)

2- **wr_en_cp,rd_en_cp,underflow_cp**
   (it is impossible for overflow to be high when there is no write operation)

3- **wr_en_cp,rd_en_cp,wr_ack_cp**
   (it is impossible for wr_ack to be high when there is no write operation)

4- **wr_en_cp,rd_en_cp,fullcp**
   (it is impossible for full to be high when there is a read operation)

Monitor

```
    import FIFO_transaction_pkg::*;
    import FIFO_scoreboard_pkg::*;
    import FIFO_coverage_pkg::*;
    import shared_pkg::*;

    module FIFO_monitor(FIFO_if.MONITOR fifoif);
        FIFO_coverage f_cov;
        FIFO_transaction f_txn_mon;
        FIFO_scoreboard f_sb;
        initial begin
            f_cov = new();
            f_txn_mon = new();
            f_sb = new();

            forever begin
                //=====wait for test to trigger event================
                wait(fifoif.ev_trig.triggered);
                @(negedge fifoif.clk);

                //=====sample signals from interface into the transaction object=======
                f_txn_mon.rst_n = fifoif.rst_n;
                f_txn_mon.rd_en = fifoif.rd_en;
                f_txn_mon.wr_en = fifoif.wr_en;
                f_txn_mon.data_in = fifoif.data_in;
                f_txn_mon.data_out = fifoif.data_out;
                f_txn_mon.wr_ack = fifoif.wr_ack;
                f_txn_mon.full = fifoif.full;
                f_txn_mon.empty = fifoif.empty;
                f_txn_mon.almostempty = fifoif.almostempty;
                f_txn_mon.almostfull = fifoif.almostfull;
                f_txn_mon.overflow = fifoif.overflow;
                f_txn_mon.underflow = fifoif.underflow;
                //=====forking to run coverage and scoreboard in parallel=========
                fork
                    begin
                        f_cov.sample_data(f_txn_mon);
                    end

                    begin
                        f_sb.check_data(f_txn_mon);
                    end
                join
                //====check if test finished then stop simulation==========
                if(test_finished) begin
                    $display("TEST FINISHED: Correct Count = %0d, Error Count = %0d",
correct_count,error_count);
                    $stop;
                end
            end
        end
    endmodule
```

Testbench:

```
import shared_pkg::*;
import FIFO_transaction_pkg::*;

module tb(FIFO_if.TEST fifoif);

    FIFO_transaction f_txn_tb;

    initial begin
        f_txn_tb = new();
        test_finished = 0;

        //====TESET RESET=============================================//
        fifoif.rst_n = 0;
        ->fifoif.ev_trig;
        repeat(10) @(negedge fifoif.clk);
        fifoif.rst_n = 1;
        //====TESET WRITE CONSECUTIVE ============================================//
        repeat(100)begin
            assert(f_txn_tb.randomize());
            fifoif.rd_en = 0;
            fifoif.wr_en = 1;
            fifoif.data_in = f_txn_tb.data_in;
            fifoif.rst_n = 1;
            -> fifoif.ev_trig;
            @(negedge fifoif.clk);
        end
        //====TESET READ CONSECUTIVE ============================================//
        repeat(100)begin
            assert(f_txn_tb.randomize());
            fifoif.rd_en = 1;
            fifoif.wr_en = 0;
            fifoif.data_in = f_txn_tb.data_in;
            fifoif.rst_n = 1;
            -> fifoif.ev_trig;
            @(negedge fifoif.clk);
        end
        //====TESET READ and WRITE CONSECUTIVE
=============================================//
        repeat(100)begin
            assert(f_txn_tb.randomize());
            fifoif.rd_en = 1;
            fifoif.wr_en = 1;
            fifoif.data_in = f_txn_tb.data_in;
            fifoif.rst_n = 1;
            -> fifoif.ev_trig;
            @(negedge fifoif.clk);
        end
        //====TESET RANDOMIZED=========================================//
        repeat(10000)begin
            assert(f_txn_tb.randomize());
            fifoif.rd_en = f_txn_tb.rd_en;
```

```
            fifoif.wr_en = f_txn_tb.wr_en;
            fifoif.data_in = f_txn_tb.data_in;
            fifoif.rst_n = f_txn_tb.rst_n;
            -> fifoif.ev_trig;
            @(negedge fifoif.clk);
        end
        //====TESET FINISHED==============================================//

        test_finished = 1;
        -> fifoif.ev_trig;
        @(negedge fifoif.clk);

    end
endmodule
```

Top module

```
module top;
    bit clk;

    initial begin
        clk = 0;
        forever begin
             #1 clk = ~clk;
        end
    end

    FIFO_if fifoif(clk);

    always_comb begin
        if(!fifoif.rst_n) begin
            assert final (
                            fifoif.empty&&
                            !fifoif.full&&
                            !fifoif.almostempty &&
                            !fifoif.almostfull &&
                            !fifoif.wr_ack &&
                            !fifoif.overflow &&
                            !fifoif.underflow &&
                            fifoif.data_out == '0
                        );
        end
    end

    FIFO DUT(fifoif);
    FIFO_monitor monitor(fifoif);
    tb test(fifoif);

endmodule
```

Do file

Code > run.do

You, 3 hours ago | 1 author (You)

```
vlib work
vlog FIFO_if.sv FIFO.sv shared_pkg.sv FIFO_transaction.sv FIFO_coverage.sv FIFO_scoreboard.sv FIFO_monitor.sv tb.sv top.sv  +cover
vsim -voptargs=+acc work.top -cover
add wave *
coverage save top.ucdb  -onexit
run 0
do wave.do
run -all
```

You, 6 hours ago • Add assertions, coverage, and monitor for FIFO …

Wave do:

Code > wave.do

You, 3 hours ago | 1 author (You)

```
onerror {resume}
quietly WaveActivateNextPane {} 0
add wave -noupdate /top/clk
add wave -noupdate -group Interface /top/fifoif/FIFO_WIDTH
add wave -noupdate -group Interface /top/fifoif/FIFO_DEPTH
add wave -noupdate -group Interface /top/fifoif/clk
add wave -noupdate -group Interface /top/fifoif/data_in
add wave -noupdate -group Interface /top/fifoif/rst_n
add wave -noupdate -group Interface /top/fifoif/wr_en
add wave -noupdate -group Interface /top/fifoif/rd_en
add wave -noupdate -group Interface /top/fifoif/data_out
add wave -noupdate -group Interface /top/fifoif/wr_ack
add wave -noupdate -group Interface /top/fifoif/full
add wave -noupdate -group Interface /top/fifoif/empty
add wave -noupdate -group Interface /top/fifoif/almostfull
add wave -noupdate -group Interface /top/fifoif/almostempty
add wave -noupdate -group Interface /top/fifoif/underflow
add wave -noupdate -group Interface /top/fifoif/overflow
add wave -noupdate -group Interface /top/fifoif/ev_trig
add wave -noupdate -expand -group {Coverage signals} /top/monitor/f_cov
add wave -noupdate -expand -group {Monitor txn} /top/monitor/f_txn_mon
add wave -noupdate -expand -group Scoreboard /top/monitor/f_sb
add wave -noupdate -group ASSERTIONS /top/DUT/wr_ack_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/overflow_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/underflow_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/empty_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/full_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/almost_full_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/almost_empty_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/wr_ptr_wraparound_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/rd_ptr_wraparound_asrt
add wave -noupdate -group ASSERTIONS /top/DUT/ptr_threshold_asrt
TreeUpdate [SetDefaultTree]
WaveRestoreCursors {{Cursor 1} {57 ns} 0}
quietly wave cursor active 1
configure wave -namecolwidth 150
configure wave -valuecolwidth 100
configure wave -justifyvalue left
configure wave -signalnamewidth 0
configure wave -snapdistance 10
configure wave -datasetprefix 0
configure wave -rowmargin 4
configure wave -childrowmargin 2
configure wave -gridoffset 0
configure wave -gridperiod 1
configure wave -griddelta 40
configure wave -timeline 0
configure wave -timelineunits ns
update
WaveRestoreZoom {0 ns} {1 us}
```

You, 3 hours ago • added double event handshake …

Questasim Snippets

```
Transcript
# Loading work.tb_sv_unit(fast)
# Loading work.tb(fast)
# ** UI-Msg: (vish-4014) No objects found matching '/top/DUT/rd_ptr_wraparound_asrt'.
# Executing ONERROR command at macro ./wave.do line 31
# TEST FINISHED: Correct Count = 10302, Error Count = 0
# ** Note: $stop    : FIFO_monitor.sv(46)
#    Time: 20620 ns  Iteration: 1  Instance: /top/monitor
# Break in Module FIFO_monitor at FIFO_monitor.sv line 46

VSIM 6>
Now: 20,620 ns  Delta: 1        sim:/top/monitor/#INITIAL#10
```

Functional coverage

Covergroups

Name	Class Type	Coverage	Goal	% of Goal	Status	Included	Merge_in
/FIFO_coverage_pkg/FIFO_coverage		100.00%					
TYPE cov_group		100.00%	100	100.00...			
CVP cov_group::wr_en_cp		100.00%	100	100.00...			
CVP cov_group::rd_en_cp		100.00%	100	100.00...			
CVP cov_group::wr_ack_cp		100.00%	100	100.00...			
CVP cov_group::fullcp		100.00%	100	100.00...			
CVP cov_group::almostfull_cp		100.00%	100	100.00...			
CVP cov_group::empty_cp		100.00%	100	100.00...			
CVP cov_group::almostempty_cp		100.00%	100	100.00...			
CVP cov_group::overflowcp		100.00%	100	100.00...			
CVP cov_group::underflow_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_wr_ack_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_full_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_empty_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_almostfull_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_almostempty_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_overflow_cp		100.00%	100	100.00...			
CROSS cov_group::wr_rd_en_underflow_cp		100.00%	100	100.00...			

Assertions | Cover Directives | Covergroups | Wave | top.sv | FIFO_monitor.sv | FIFO.sv

Assertions

Name	Assertion Type	Language	Enable	Failure Count	Pass Count	Active Count	Memory	Peak Memory
/top/#ublk#31584#14/immed__15	Immediate	SVA	on	0	1	-	-	-
/top/DUT/wr_ack_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/overflow_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/underflow_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/empty_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/full_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/almost_full_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/almost_empty_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/wr_ptr_wraparound_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/ptr_threshold_asrt	Concurrent	SVA	on	0	1	-	0B	0B
/top/DUT/#ublk#306607#85/immed__86	Immediate	SVA	on	0	1	-	-	-
/top/test/#ublk#1954#18/immed__19	Immediate	SVA	on	0	1	-	-	-
/top/test/#ublk#1954#28/immed__29	Immediate	SVA	on	0	1	-	-	-
/top/test/#ublk#1954#38/immed__39	Immediate	SVA	on	0	1	-	-	-
/top/test/#ublk#1954#48/immed__49	Immediate	SVA	on	0	1	-	-	-

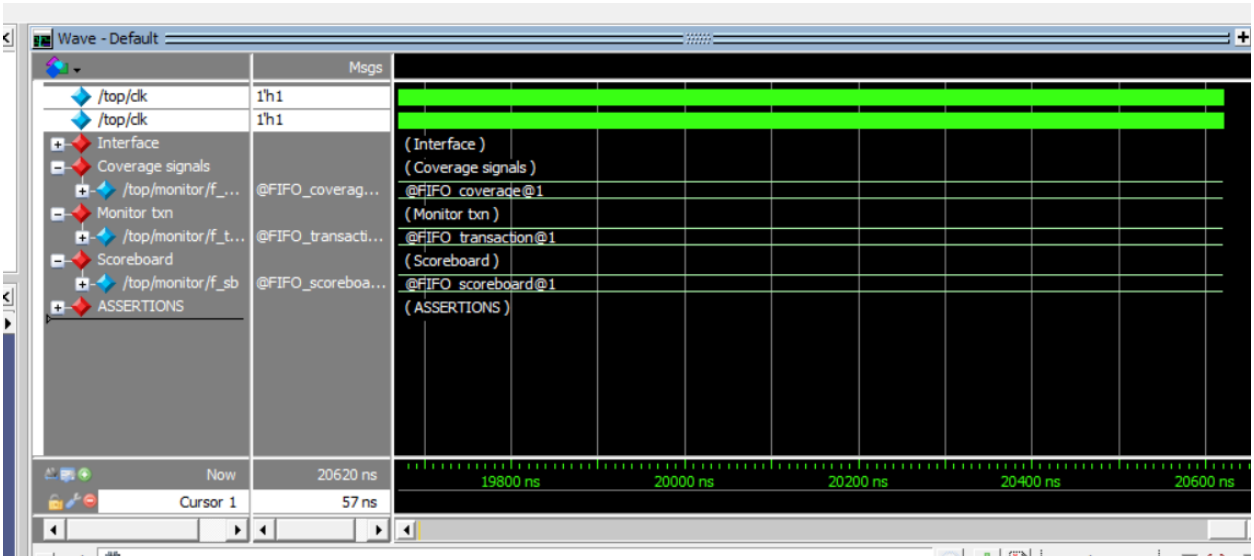
Assertions | Cover Directives | Covergroups | Wave | top.sv | FIFO_monitor.sv | FIFO.sv

Cover Directives

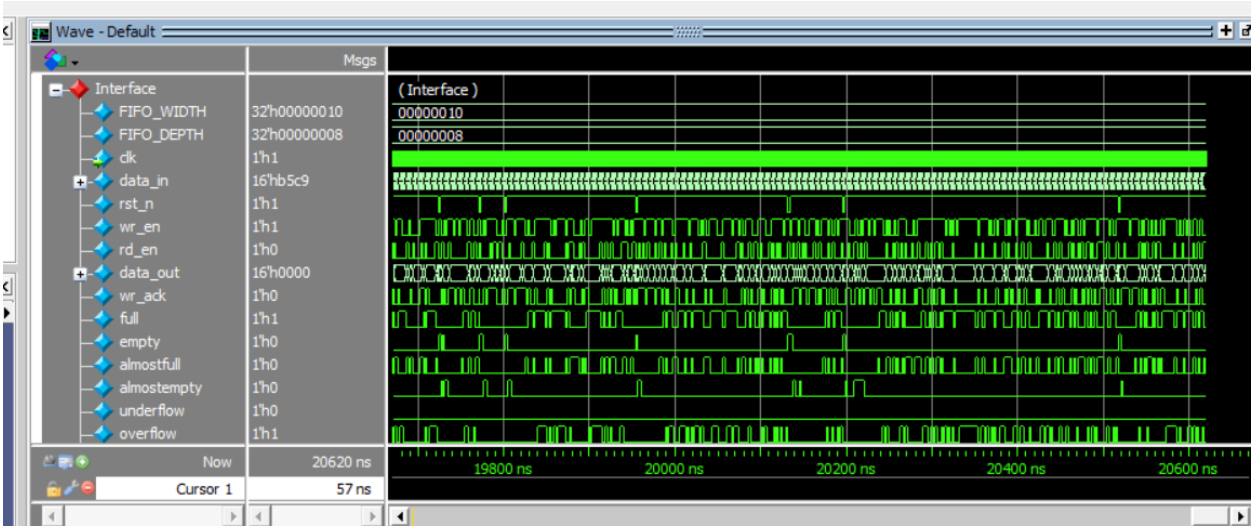
Name	Language	Enabled	Log	Count	AtLeast	Limit	Weight	Cmplt %	Cmplt graph	Included
/top/DUT/wr_ack_cov	SVA	✓	Off	4103	1	Unlimited	1	100%		✓
/top/DUT/overflow_cov	SVA	✓	Off	2955	1	Unlimited	1	100%		✓
/top/DUT/underflow_cov	SVA	✓	Off	178	1	Unlimited	1	100%		✓
/top/DUT/empty_cov	SVA	✓	Off	370	1	Unlimited	1	100%		✓
/top/DUT/full_cov	SVA	✓	Off	4196	1	Unlimited	1	100%		✓
/top/DUT/almost_full_cov	SVA	✓	Off	2653	1	Unlimited	1	100%		✓
/top/DUT/almost_empty_cov	SVA	✓	Off	528	1	Unlimited	1	100%		✓
/top/DUT/wr_ptr_wraparound_cov	SVA	✓	Off	435	1	Unlimited	1	100%		✓
/top/DUT/ptr_threshold_cov	SVA	✓	Off	10119	1	Unlimited	1	100%		✓

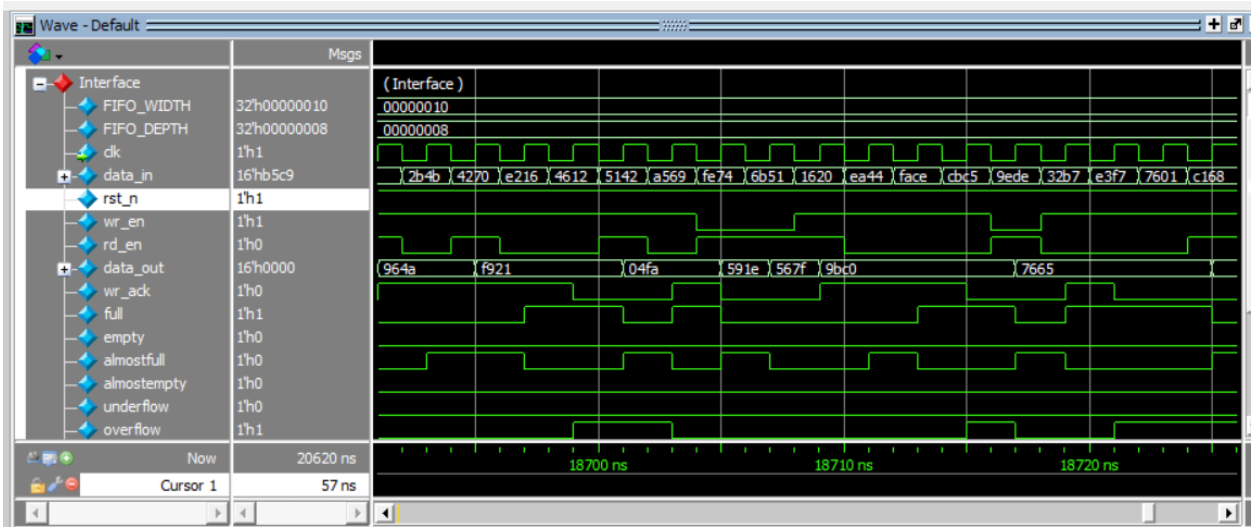
Waveform:

(don't mind the duplicate clk, too lazy to remove it)

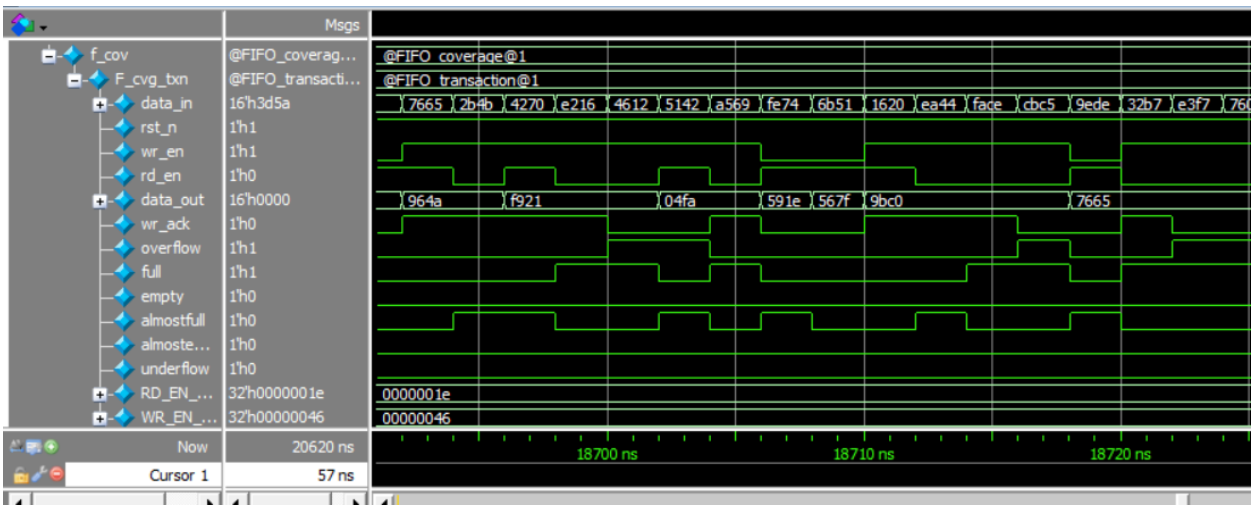


Interface signals

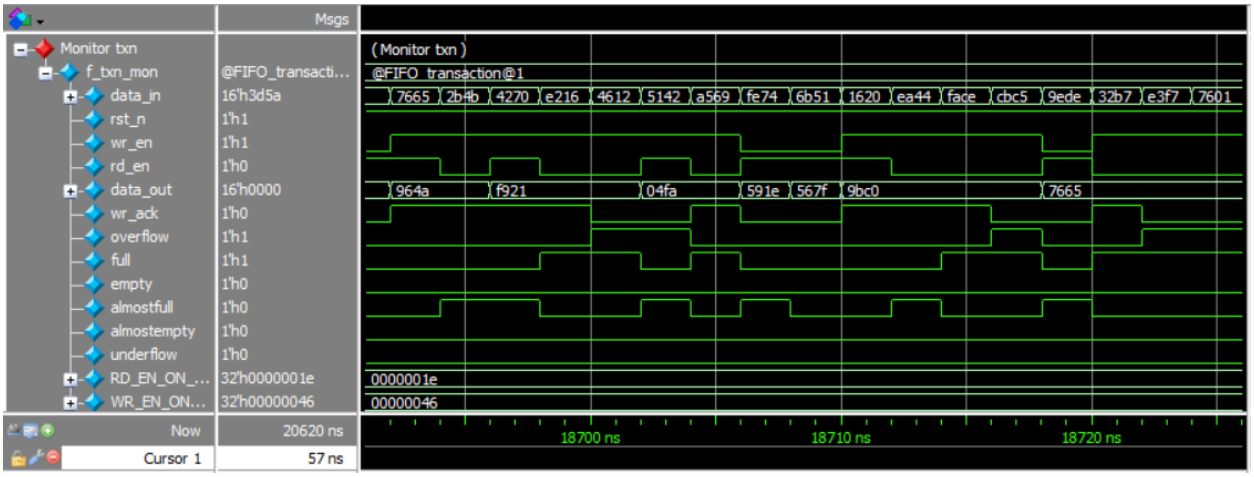




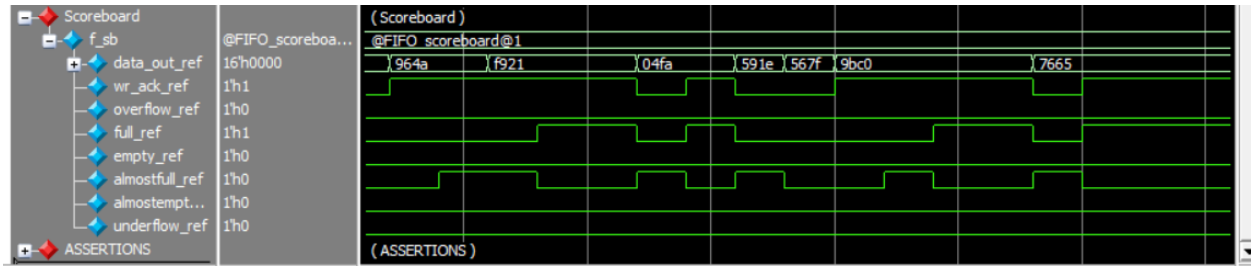
Coverage transaction object



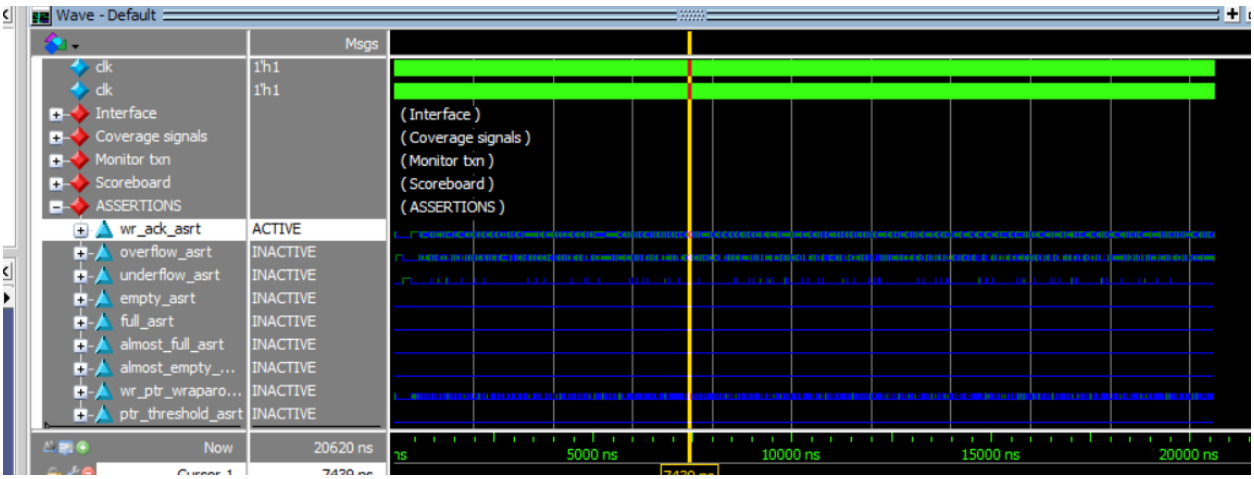
Monitor Transaction object



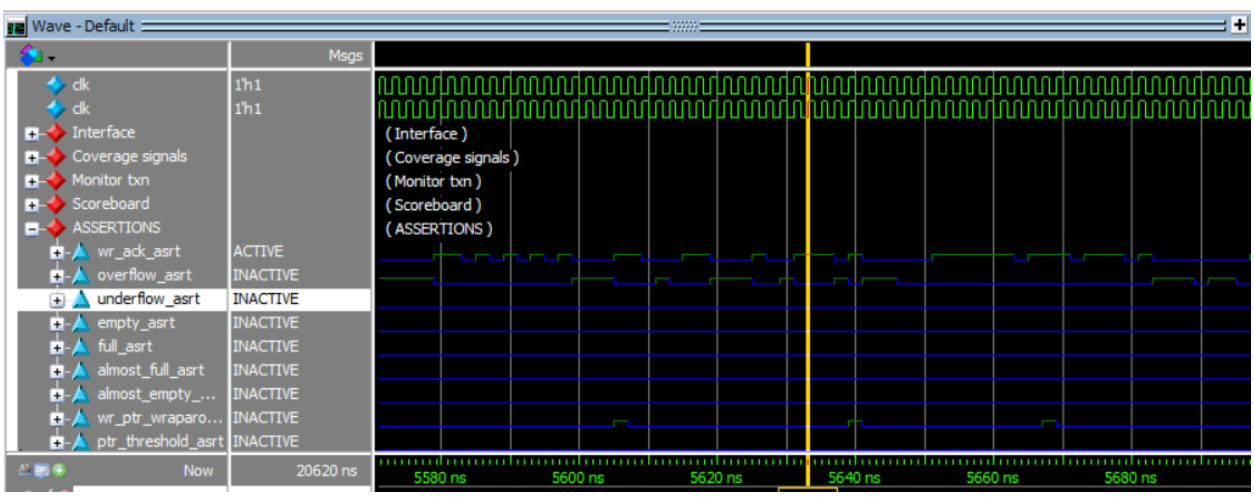
Scoreboard reference model out signals



Assertions waves



A little zoom in



Coverage Report:

Code coverage:

Toggle:

```
================================================================================
=== Instance: /top/fifoif
=== Design Unit: work.FIFO_if
================================================================================
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         86        86         0   100.00%

==============================Toggle Details==============================

Toggle Coverage for instance /top/fifoif --

                                          Node      1H->0L      0L->1H  "Coverage"
                                          ---------------------------------------
                                   almostempty           1           1      100.00
                                    almostfull           1           1      100.00
                                           clk           1           1      100.00
                                  data_in[15-0]          1           1      100.00
                                 data_out[15-0]          1           1      100.00
                                         empty           1           1      100.00
                                          full           1           1      100.00
                                      overflow           1           1      100.00
                                         rd_en           1           1      100.00
                                         rst_n           1           1      100.00
                                     underflow           1           1      100.00
                                        wr_ack           1           1      100.00
                                         wr_en           1           1      100.00

Total Node Count     =         43
Toggled Node Count   =         43
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (86 of 86 bins)

================================================================================
```

```
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         20        20         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top/DUT --

                                          Node      1H->0L      0L->1H  "Coverage"
                                          ---------------------------------------
                                     count[3-0]         1           1      100.00
                                    rd_ptr[2-0]         1           1      100.00
                                    wr_ptr[2-0]         1           1      100.00

Total Node Count     =         10
Toggled Node Count   =         10
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (20 of 20 bins)
```

Branch:

```
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        27        27         0   100.00%

================================Branch Details================================

Branch Coverage for instance /top/DUT

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File FIFO.sv
------------------------------------IF Branch------------------------------------
    19                                     10489     Count coming in to IF
    19             1                         370         if (!fifoif.rst_n) begin

    24             1                        4161         else if (fifoif.wr_en && count < fifoif.FIFO_DEPTH) begin

    30             1                        5958         else begin

Branch totals: 3 hits of 3 branches = 100.00%

------------------------------------IF Branch------------------------------------
    32                                      5958     Count coming in to IF
    32             1                        3016             if (fifoif.full & fifoif.wr_en)

    34             1                        2942             else

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    40                                     10489     Count coming in to IF
    40             1                         370         if (!fifoif.rst_n) begin

    45             1                        2980         else if (fifoif.rd_en && count != 0) begin

    50             1                        7139         else begin  //BUG: handling of underflow was missing

Branch totals: 3 hits of 3 branches = 100.00%

------------------------------------IF Branch------------------------------------
    51                                      7139     Count coming in to IF
    51             1                         179             if (fifoif.empty & fifoif.rd_en)

    53             1                        6960             else

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    59                                      8553     Count coming in to IF
    59             1                         361         if (!fifoif.rst_n) begin

    63             1                        8192         else begin

Branch totals: 2 hits of 2 branches = 100.00%
```

```
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    64                                     8192     Count coming in to IF
    64             1                       2835             if  ( ({fifoif.wr_en, fifoif.rd_en} == 2'b10) && !fifoif.full)

    66             1                        836             else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b01) && !fifoif.empty)

    69             1                         62             else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b11) && fifoif.empty)

    72             1                        880             else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b11) && fifoif.full)

                                            3579     All False Count
Branch totals: 5 hits of 5 branches = 100.00%

------------------------------------IF Branch------------------------------------
    77                                     4791     Count coming in to IF
    77             1                       1322     assign fifoif.full = (count == fifoif.FIFO_DEPTH)? 1 : 0;

    77             2                       3469     assign fifoif.full = (count == fifoif.FIFO_DEPTH)? 1 : 0;

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    78                                     4791     Count coming in to IF
    78             1                        207     assign fifoif.empty = (count == 0)? 1 : 0;

    78             2                       4584     assign fifoif.empty = (count == 0)? 1 : 0;

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    80                                     4791     Count coming in to IF
    80             1                       1595     assign fifoif.almostfull = (count == fifoif.FIFO_DEPTH-1)? 1 : 0; //BUG: was written as fifodepth -2 instead of 1

    80             2                       3196     assign fifoif.almostfull = (count == fifoif.FIFO_DEPTH-1)? 1 : 0; //BUG: was written as fifodepth -2 instead of 1

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    81                                     4791     Count coming in to IF
    81             1                        249     assign fifoif.almostempty = (count == 1)? 1 : 0;

    81             2                       4542     assign fifoif.almostempty = (count == 1)? 1 : 0;

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    85                                     6413     Count coming in to IF
    85             1                        357             if(!fifoif.rst_n) begin

                                            6056     All False Count
Branch totals: 2 hits of 2 branches = 100.00%
```

Statement Coverage:

```
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                      31        31         0   100.00%

================================Statement Details================================

Statement Coverage for instance /top/DUT --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File FIFO.sv
    8                                                module FIFO(FIFO_if.DUT fifoif);

    9

    10

    11                                               localparam max_fifo_addr = $clog2(fifoif.FIFO_DEPTH);

    12

    13                                               reg [fifoif.FIFO_WIDTH-1:0] mem [fifoif.FIFO_DEPTH-1:0];

    14

    15                                               reg [max_fifo_addr-1:0] wr_ptr, rd_ptr;

    16                                               reg [max_fifo_addr:0] count;

    17

    18             1                         10489   always @(posedge fifoif.clk or negedge fifoif.rst_n) begin

    19                                                 if (!fifoif.rst_n) begin

    20             1                           370         wr_ptr <= 0;

    21             1                           370         fifoif.wr_ack <= 0;     //BUG: reset wr_ack was missing

    22             1                           370         fifoif.overflow <= 0;   //BUG: reset for overflow was missing

    23                                                 end

    24                                                 else if (fifoif.wr_en && count < fifoif.FIFO_DEPTH) begin

    25             1                          4161         mem[wr_ptr] <= fifoif.data_in;

    26             1                          4161         fifoif.wr_ack <= 1;

    27             1                          4161         wr_ptr <= wr_ptr + 1;
```

Functional Coverage:

```
================================================================================
=== Instance: /FIFO_coverage_pkg
=== Design Unit: work.FIFO_coverage_pkg
================================================================================

Covergroup Coverage:
    Covergroups                    1        na        na   100.00%
        Coverpoints/Crosses       16        na        na        na
            Covergroup Bins       66        66         0   100.00%
----------------------------------------------------------------------------------------------------
Covergroup                                          Metric       Goal       Bins    Status

----------------------------------------------------------------------------------------------------
 TYPE /FIFO_coverage_pkg/FIFO_coverage/cov_group    100.00%        100          -    Covered
    covered/total bins:                                  66         66          -
    missing/total bins:                                   0         66          -
    % Hit:                                           100.00%        100          -
    Coverpoint wr_en_cp                              100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     2998          1          -    Covered
        bin auto[1]                                     7303          1          -    Covered
    Coverpoint rd_en_cp                              100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     7076          1          -    Covered
        bin auto[1]                                     3225          1          -    Covered
    Coverpoint wr_ack_cp                             100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     6141          1          -    Covered
        bin auto[1]                                     4161          1          -    Covered
    Coverpoint fullcp                                100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     6025          1          -    Covered
        bin auto[1]                                     4277          1          -    Covered
    Coverpoint almostfull_cp                         100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     7602          1          -    Covered
        bin auto[1]                                     2700          1          -    Covered
    Coverpoint empty_cp                              100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     9927          1          -    Covered
        bin auto[1]                                      375          1          -    Covered
    Coverpoint almostempty_cp                        100.00%        100          -    Covered
        covered/total bins:                                2          2          -
```

```
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     9769          1          -    Covered
        bin auto[1]                                      533          1          -    Covered
    Coverpoint overflowcp                            100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                     7286          1          -    Covered
        bin auto[1]                                     3016          1          -    Covered
    Coverpoint underflow_cp                          100.00%        100          -    Covered
        covered/total bins:                                2          2          -
        missing/total bins:                                0          2          -
        % Hit:                                       100.00%        100          -
        bin auto[0]                                    10123          1          -    Covered
        bin auto[1]                                      179          1          -    Covered
```

```
            bin auto[1]                                   179          1          -    Covered
        Cross wr_rd_en_wr_ack_cp                       100.00%        100          -    Covered
            covered/total bins:                              6          6          -
            missing/total bins:                              0          6          -
            % Hit:                                     100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>               1326          1          -    Covered
                bin <auto[1],auto[0],auto[1]>               2835          1          -    Covered
                bin <auto[1],auto[1],auto[0]>                927          1          -    Covered
                bin <auto[1],auto[0],auto[0]>               2215          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                972          1          -    Covered
                bin <auto[0],auto[0],auto[0]>               2026          1          -    Covered
            Illegal and Ignore Bins:
                illegal_bin wr_en0_rd_en1_wr_ack1              0                     -    ZERO
                illegal_bin wr_en0_rd_en0_wr_ack1              0                     -    ZERO
        Cross wr_rd_en_full_cp                         100.00%        100          -    Covered
            covered/total bins:                              6          6          -
            missing/total bins:                              0          6          -
            % Hit:                                     100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[0]>               2253          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                972          1          -    Covered
                bin <auto[1],auto[0],auto[1]>               3458          1          -    Covered
                bin <auto[1],auto[0],auto[0]>               1592          1          -    Covered
                bin <auto[0],auto[0],auto[1]>                819          1          -    Covered
                bin <auto[0],auto[0],auto[0]>               1207          1          -    Covered
            Illegal and Ignore Bins:
                illegal_bin wr_en1_rd_en1_full1                0                     -    ZERO
                illegal_bin wr_en0_rd_en1_full1                0                     -    ZERO
        Cross wr_rd_en_empty_cp                        100.00%        100          -    Covered
            covered/total bins:                              8          8          -
            missing/total bins:                              0          8          -
            % Hit:                                     100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>                 47          1          -    Covered
                bin <auto[0],auto[1],auto[1]>                165          1          -    Covered
                bin <auto[1],auto[0],auto[1]>                 79          1          -    Covered
                bin <auto[0],auto[0],auto[1]>                 83          1          -    Covered
                bin <auto[1],auto[1],auto[0]>               2206          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                807          1          -    Covered
                bin <auto[1],auto[0],auto[0]>               4971          1          -    Covered
                bin <auto[0],auto[0],auto[0]>               1943          1          -    Covered
        Cross wr_rd_en_almostfull_cp                   100.00%        100          -    Covered
            covered/total bins:                              8          8          -
            missing/total bins:                              0          8          -
            % Hit:                                     100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>               1427          1          -    Covered
                bin <auto[0],auto[1],auto[1]>                361          1          -    Covered
                bin <auto[1],auto[0],auto[1]>                354          1          -    Covered
                bin <auto[0],auto[0],auto[1]>                558          1          -    Covered
                bin <auto[1],auto[1],auto[0]>                826          1          -    Covered
```

```
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>                1427          1          -    Covered
                bin <auto[0],auto[1],auto[1]>                 361          1          -    Covered
                bin <auto[1],auto[0],auto[1]>                 354          1          -    Covered
                bin <auto[0],auto[0],auto[1]>                 558          1          -    Covered
                bin <auto[1],auto[1],auto[0]>                 826          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                 611          1          -    Covered
                bin <auto[1],auto[0],auto[0]>                4696          1          -    Covered
                bin <auto[0],auto[0],auto[0]>                1468          1          -    Covered
        Cross wr_rd_en_almostempty_cp                     100.00%        100          -    Covered
            covered/total bins:                                 8          8          -
            missing/total bins:                                 0          8          -
            % Hit:                                        100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>                 253          1          -    Covered
                bin <auto[0],auto[1],auto[1]>                  42          1          -    Covered
                bin <auto[1],auto[0],auto[1]>                 145          1          -    Covered
                bin <auto[0],auto[0],auto[1]>                  93          1          -    Covered
                bin <auto[1],auto[1],auto[0]>                2000          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                 930          1          -    Covered
                bin <auto[1],auto[0],auto[0]>                4905          1          -    Covered
                bin <auto[0],auto[0],auto[0]>                1933          1          -    Covered
        Cross wr_rd_en_overflow_cp                        100.00%        100          -    Covered
            covered/total bins:                                 6          6          -
            missing/total bins:                                 0          6          -
            % Hit:                                        100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>                 880          1          -    Covered
                bin <auto[1],auto[0],auto[1]>                2136          1          -    Covered
                bin <auto[1],auto[1],auto[0]>                1373          1          -    Covered
                bin <auto[1],auto[0],auto[0]>                2914          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                 972          1          -    Covered
                bin <auto[0],auto[0],auto[0]>                2026          1          -    Covered
            Illegal and Ignore Bins:
                illegal_bin wr_en0_rd_en1_ov1                   0                     -    ZERO
                illegal_bin wr_en0_rd_en0_ov1                   0                     -    ZERO
        Cross wr_rd_en_underflow_cp                       100.00%        100          -    Covered
            covered/total bins:                                 6          6          -
            missing/total bins:                                 0          6          -
            % Hit:                                        100.00%        100          -
            Auto, Default and User Defined Bins:
                bin <auto[1],auto[1],auto[1]>                  62          1          -    Covered
                bin <auto[1],auto[1],auto[0]>                2191          1          -    Covered
                bin <auto[0],auto[1],auto[1]>                 117          1          -    Covered
                bin <auto[0],auto[1],auto[0]>                 855          1          -    Covered
                bin <auto[1],auto[0],auto[0]>                5050          1          -    Covered
                bin <auto[0],auto[0],auto[0]>                2026          1          -    Covered
            Illegal and Ignore Bins:
                illegal_bin wr_en1_rd_en0_uf1                   0                     -    ZERO
                illegal_bin wr_en0_rd_en0_uf1                   0                     -    ZERO
```

Assertion Coverage:

```
DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------------------
Name                                    Design Design   Lang File(Line)      Hits Status
                                        Unit   UnitType
--------------------------------------------------------------------------------------------
/top/DUT/wr_ack_cov                     FIFO   Verilog  SVA  FIFO.sv(138)    4103 Covered
/top/DUT/overflow_cov                   FIFO   Verilog  SVA  FIFO.sv(139)    2955 Covered
/top/DUT/underflow_cov                  FIFO   Verilog  SVA  FIFO.sv(140)     178 Covered
/top/DUT/empty_cov                      FIFO   Verilog  SVA  FIFO.sv(141)     370 Covered
/top/DUT/full_cov                       FIFO   Verilog  SVA  FIFO.sv(142)    4196 Covered
/top/DUT/almost_full_cov                FIFO   Verilog  SVA  FIFO.sv(143)    2653 Covered
/top/DUT/almost_empty_cov               FIFO   Verilog  SVA  FIFO.sv(144)     528 Covered
/top/DUT/wr_ptr_wraparound_cov          FIFO   Verilog  SVA  FIFO.sv(145)     435 Covered
/top/DUT/ptr_threshold_cov              FIFO   Verilog  SVA  FIFO.sv(146)    10119 Covered

TOTAL DIRECTIVE COVERAGE: 100.00%  COVERS: 9

ASSERTION RESULTS:
--------------------------------------------------------------------
Name                 File(Line)                  Failure      Pass
                                                 Count        Count
--------------------------------------------------------------------
/top/#ublk#31584#14/immed__15
                     top.sv(16)                      0          1
/top/DUT/#ublk#306607#85/immed__86
                     FIFO.sv(86)                     0          1
/top/DUT/wr_ack_asrt FIFO.sv(119)                    0          1
/top/DUT/overflow_asrt
                     FIFO.sv(121)                    0          1
/top/DUT/underflow_asrt
                     FIFO.sv(123)                    0          1
/top/DUT/empty_asrt  FIFO.sv(125)                    0          1
/top/DUT/full_asrt   FIFO.sv(127)                    0          1
/top/DUT/almost_full_asrt
                     FIFO.sv(129)                    0          1
/top/DUT/almost_empty_asrt
                     FIFO.sv(131)                    0          1
/top/DUT/wr_ptr_wraparound_asrt
                     FIFO.sv(133)                    0          1
/top/DUT/ptr_threshold_asrt
                     FIFO.sv(135)                    0          1
/top/test/#ublk#1954#18/immed__19
                     tb.sv(19)                       0          1
/top/test/#ublk#1954#28/immed__29
                     tb.sv(29)                       0          1
/top/test/#ublk#1954#38/immed__39
                     tb.sv(39)                       0          1
/top/test/#ublk#1954#48/immed__49
                     tb.sv(49)                       0          1
```

```
Assertion Coverage:
    Assertions                      1         1         0   100.00%
--------------------------------------------------------------------
Name                 File(Line)                  Failure      Pass
                                                 Count        Count
--------------------------------------------------------------------
/top/#ublk#31584#14/immed__15
                     top.sv(16)                      0          1
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                         2         2         0   100.00%
```

```
    ================================================================================
    === Instance: /top/test
    === Design Unit: work.tb
    ================================================================================

    Assertion Coverage:
        Assertions                       4         4         0   100.00%
    ----------------------------------------------------------------------
    Name                 File(Line)                    Failure      Pass
                                                       Count        Count
    ----------------------------------------------------------------------
    /top/test/#ublk#1954#18/immed__19
                         tb.sv(19)                           0          1
    /top/test/#ublk#1954#28/immed__29
                         tb.sv(29)                           0          1
    /top/test/#ublk#1954#38/immed__39
                         tb.sv(39)                           0          1
    /top/test/#ublk#1954#48/immed__49
                         tb.sv(49)                           0          1
    Statement Coverage:
        Enabled Coverage              Bins      Hits    Misses  Coverage
        ----------------              ----      ----    ------  --------
        Statements                      38        38         0   100.00%
```

```
    ================================================================================
    === Instance: /top/DUT
    === Design Unit: work.FIFO
    ================================================================================

    Assertion Coverage:
        Assertions                      10        10         0   100.00%
    ----------------------------------------------------------------------
    Name                 File(Line)                    Failure      Pass
                                                       Count        Count
    ----------------------------------------------------------------------
    /top/DUT/#ublk#306607#85/immed__86
                         FIFO.sv(86)                         0          1
    /top/DUT/wr_ack_asrt FIFO.sv(119)                        0          1
    /top/DUT/overflow_asrt
                         FIFO.sv(121)                        0          1
    /top/DUT/underflow_asrt
                         FIFO.sv(123)                        0          1
    /top/DUT/empty_asrt  FIFO.sv(125)                        0          1
    /top/DUT/full_asrt   FIFO.sv(127)                        0          1
    /top/DUT/almost_full_asrt
                         FIFO.sv(129)                        0          1
    /top/DUT/almost_empty_asrt
                         FIFO.sv(131)                        0          1
    /top/DUT/wr_ptr_wraparound_asrt
                         FIFO.sv(133)                        0          1
    /top/DUT/ptr_threshold_asrt
                         FIFO.sv(135)                        0          1
    Branch Coverage:
        Enabled Coverage              Bins      Hits    Misses  Coverage
        ----------------              ----      ----    ------  --------
        Branches                        27        27         0   100.00%

    ------------------------------Branch Details------------------------------
```